

Second Edition

RESEARCH METHODOLOGY IN APPLIED ECONOMICS

COLORADO STATE UNIVERSITY LIBRARY

Don E. Ethridge, Ph.D., is Chairman and Professor, Department of Agricultural and Applied Economics, Texas Tech University, Lubbock.

©2004 Blackwell Publishing
All rights reserved

Blackwell Publishing Professional
2121 State Avenue, Ames, Iowa 50014, USA

Orders: 1-800-862-6657
Office: 1-515-292-0140
Fax: 1-515-292-3348
Web site: www.blackwellprofessional.com

Blackwell Publishing Ltd
9600 Garsington Road, Oxford OX4 2DQ, UK
Tel.: +44 (0)1865 776868

Blackwell Publishing Asia
550 Swanston Street, Carlton, Victoria 3053, Australia
Tel.: +61 (0)3 8359 1011

Authorization to photocopy items for internal or personal use, or the internal or personal use of specific clients, is granted by Blackwell Publishing, provided that the base fee of \$.10 per copy is paid directly to the Copyright Clearance Center, 222 Rosewood Drive, Danvers, MA 01923. For those organizations that have been granted a photocopy license by CCC, a separate system of payments has been arranged. The fee code for users of the Transactional Reporting Service is 0-8138-2994-1/2004 \$.10.

Printed on acid-free paper in the United States of America

First edition, © 1995, Iowa State University Press
Second edition, © 2004

Library of Congress Cataloging-in-Publication Data

Ethridge, Don E. (Don Erwin)

Research methodology in applied economics: organizing, planning, and conducting economic research/Don Ethridge—2nd ed.

p. cm.

Includes bibliographical references and index.

ISBN 0-8138-2994-1 (alk. paper)

1. Economics—Research. 2. Economics—Methodology. I. Title.

HB74.5.E84 2004

330'.9072—dc22

2004001461

The last digit is the print number: 9 8 7 6 5 4 3 2 1

2

Research and Methodology

Wisdom is not a product of schooling but of the lifelong attempt to acquire it.

Albert Einstein

The term *research* evokes images of activities that to many are often incomplete and in some cases misinformed. This happens with many specialized activities. Many people envision someone engaged in a laboratory, surrounded by test tubes, compounds, or objects being studied, or in a library, hidden by volumes of documents piled around the researcher. While both activities represent valid research functions, neither represents a comprehensive perspective of the research process, particularly in economics. The first may represent some realistic impression of laboratory research in chemistry or biology. The second may provide a valid impression of research activity in literature or history.

This chapter establishes some basic definitions and distinctions among some terms that are central to the study of research methodology in economics. It also provides some classifications of research and a characterization of the research process. The chapter is divided into five sections. The first section defines and discusses research. The second provides some classifications of types of research. The third defines and discusses methodology, making a distinction between methodology and methods. The fourth presents a description of the research process, and the last section emphasizes creativity and its role in the process.

RESEARCH DEFINED AND DESCRIBED

Many authors have written about research. Definitions provided for research, while not contradictory with one another, differ substantially. *Webster's Collegiate Dictionary* (1977) defines research as "studious inquiry or examination; esp.: investigation or experimentation aimed at the discovery and interpretation of facts, revision of accepted theories or laws in the light of new facts, or practical application of such new or revised theories or laws." Andrew and Hildebrand

(1982, p. 3) define research as "the orderly procedure by which man increases his knowledge." Ghebremedhin and Tweeten (1994, p. 4) define it as applying "the scientific method to study hypothetical propositions of presumed relations among phenomena." Leedy (1985, p. 4) says research is "the manner in which we attempt to solve problems in a systematic effort to push back the frontiers of human ignorance or to confirm the validity of the solutions to problems others have presumably resolved." In many cases (e.g., Committee on Science, Engineering, and Public Policy, 1995), authors discuss and attempt to describe research without trying to define it.

While each of these definitions makes a meaningful statement about research, the following is offered as an accurate and comprehensive, yet succinct, definition:

Research is the systematic approach to obtaining and confirming new and reliable knowledge.

This definition identifies several characteristics that are essential for an accurate and comprehensive definition. First, research is not limited to certain types of activities such as laboratory methods or literature searches. Additionally, the definition indicates that research is systematic and orderly.¹ As an approach, research follows a sequence or a series of steps (with variations in the components and their order). Lastly, its purpose is new knowledge (what is not already known), and the knowledge must be reliable (demonstrable to others based on reason or evidence). The emphasis is on both *new* and *reliable*, although both are not always dealt with in the same individual effort. For example, some research may focus entirely on establishing reliability (i.e., *confirming*) of things we think we know.

While our specific interest is research in economics and thus we are interested in a definition of research that is workable for economics, the above definition is applicable irrespective of the discipline. We must, however, understand the similarities and differences in how the research process is conducted in economics and other disciplines. Even within economics, research can be done in many different ways. It is also useful to understand how other disciplines may validly differ, if only to keep ourselves from becoming inflexible in recognizing avenues to reliable knowledge. Therefore, both differences and similarities with other disciplines will be periodically acknowledged.

Note that the term *truth* was not used in the definition of research. Some authors (e.g., Larrabee, 1964) use the word, but we will avoid it within the context of this book. Truth implies something known with certainty, something beyond questioning. This concept of truth is outside the productive realm of thinking by researchers. Truth is a luxury we may allow ourselves in our private beliefs, but is a potentially deadly attitude in research. To "accept" an outcome as either impossible or certain is to deny the possibility of that outcome being different from

what we think we already know. The statement attributed to Will Rogers, a twentieth-century American humorist and philosopher, reflects the spirit of this attitude: "It ain't what we don't know that gets us in trouble; it's what we know that ain't so." Or, in the words of Nobel physicist Richard Feynman (1999, p. 248):

To make progress in understanding, we must remain modest and allow that we do not know. Nothing is certain or proved beyond all doubt. . . . And as you develop more information . . . , it is not that you are finding out the truth, but that you are finding out that this or that is more or less likely.

When we use *truth* in a research setting, it is with a limited meaning. This will be addressed in more detail later in the book.

When describing research, it may be useful to specify what it is not as well as what it is. Research is *not*:

A. *Accidental discovery*, although new knowledge may result from accidental discovery, and accidental discovery may occur within a structured research process. Accidental discovery, when it occurs, usually takes the form of a phenomenon that has not been previously noticed. Such a discovery or observation *may* lead to a structured research process that is directed toward verifying or understanding the observation. That process may in turn lead to new reliable knowledge.

B. *Data collection*, although gathering data is frequently a part of research. The data themselves may not constitute, in the economist's perspective, "reliable knowledge." The data are an intermediate step toward knowledge, which entail relationships among forces, factors, or variables. We should acknowledge that gathering data in some fields (e.g., applied statistics) may constitute the research process; however, gathering data so that it constitutes a statistically valid sample of a population is an objective. Other disciplines, including the laboratory sciences (chemistry, biology, etc.) and some field sciences that attempt to apply the controlled experimental design to situations without all conditions controlled may be concerned primarily with producing valid experimental data, which are used to test a hypothesis, typically of a simple causal relationship. To the economist, those data may provide the means to achieve another objective. Examples include the examination and evaluation of more complex relationships and testing hypotheses about them.

C. *Searching out published research results in libraries*, although this is an important part of the research process. The point is that the literature search in economics is an important early step, but research does not stop there. The research process always includes synthesis and analysis, even if it is directed at assessment of the prior studies. Some disciplines (e.g., literature and history) rely more heavily on synthesis and analysis of prior studies as the major component of the research process than does economics. The general public's perception that research consists

merely of reviewing the literature is misleading because that process, by itself, may produce no new knowledge. Producing new knowledge from prior literature alone must involve, at a minimum, synthesis of the prior material.

Research is:

A. *Searching for explanations* of events, phenomena, relationships, and causes, at least in economic research. Some research is occupied primarily with estimation or determination of relatively simple (e.g., two-variable) relationships—*what* occurs when something else happens. We usually want to know, in addition, *how* it occurs or *why* it occurs. Other research addresses more complex relationships, often dealing with interactions among the simpler relationships. Both are important. We often must grasp the simpler relationships before progressing to the more complex ones.

B. *A process.* As such, research is also planned and managed. Unless structured carefully, the research knowledge cannot be reliable (demonstrable to others on the basis of generally accepted rules of evidence). As a process, it also is usually interactive. Research is not done in isolation from prior research or stimulation from peers and coworkers. The process is also creative. From one perspective, we are creating new understanding. From another perspective, each research activity (project) is in some respects unique and requires a design specific to that activity. Research also is circular—it always leads to more research questions.

A distinction between research as it relates to physical or biological phenomena and research as it relates to social phenomena, of which economic research is a part, is useful. Because the nature of what is being studied in each kind of research has different characteristics with different capacities for direct observability/measurability, each demands some variations in approach to extending human knowledge about them (Committee on Science, Engineering, and Public Policy, 1995, p. 3). In addition, there seems to be something vague and mysterious about economic research. Perhaps most people intuitively know that they do not grasp what the activity includes. This contrasts with a public perception, correct or not, that people have a realistic grasp of the role and functioning of other research fields such as medical research or engineering research. Embodied within this attitude may be the notion that economic research is an activity or exercise confined to academia and a few government agencies or foundations. Even more damaging is the perception that economic research is without practical application: that is, it is not “useful.”

We maintain that these perceptions are without basis. *All* well-designed and conducted economic research has potential application. The applications may not be immediately obvious in highly theoretical, “basic” research and may be more obvious in more “applied” research efforts. One must realize, however, that the applied research is usually dependent on certain basic research being or having

already been done, so that the basic research capital stock must be maintained to produce the applied results. Also, the two often progress hand in hand and are difficult to distinguish. Well-developed and executed research, whether basic or applied, focused on defined issues always has application toward relevant problems or issues by persons in the discipline or by public and/or private decision makers. The failure to see its applications has two primary sources: (1) some users of research cannot easily grasp the applications of research because they are not formally trained or experienced in the specialized methods of economic research and reasoning; and (2) researchers often do not provide adequate interpretations and guidance on application (use) of the research to the potential users of their research. It is the responsibility of researchers to help users of their research to understand its implications and to help ensure that the research is used effectively and not to draw invalid implications.

Research takes place in the private sector as well as the public sector. The greater proportion of *published* research is associated with researchers in universities and government, where widespread dissemination of research procedures, results, and conclusions is regarded as an important part of the institutional missions. Many business firms, industry organizations, and consulting firms, for example, likewise conduct research. There *may* be a tendency for research conducted in the public sector to be more rigorous and objective because it tends to be subject to greater scrutiny, but the contention is at best an untested hypothesis. This does not suggest that any research result is more or less reliable on the basis of its source. But "research" by a firm that evaluates the benefits of a product in which the firm has a financial interest may need to be examined with greater caution than an evaluation by a group with no stake in the outcome. Nevertheless, much research is conducted within the private sector, and the validity of that research is as important as that in the public sector.

CLASSIFICATIONS OF RESEARCH

The characterization of thought, actions, and processes into classifications or divisions of general types helps us make distinctions. Before we can conceptualize, we must first *define* and then *classify* (Piaget, 1954, p. 357). Definitions and classifications are addressed throughout this book because they are fundamental to understanding the subject matter. Their necessity is explained by Bloom et al. (1956). Knowledge of various levels and types fosters intellectual abilities or skills. The hierarchy of knowledge runs from knowledge of specifics (including terminology and facts), to knowledge of dealing with facts (including classifications, methods, and methodology), to knowledge of universals and abstractions in a field (including principles, generalizations, theories, and structures). Intellectual abilities, while not completely dependent on knowledge, influence one's mastery of knowledge. The spectrum of intellectual abilities and skills progresses from comprehension (translation, interpretation, and extrapolation), to application. It then progresses to

analysis (of elements, relationships, and organizational principles), to synthesis (of communication, plans, or abstract relations), and to evaluation.

In classifying research, there are different criteria on which to base the classifications, and they may each be useful for specific purposes. A brief discussion of several of them is provided below. These classifications are introduced here, but several will be discussed again in subsequent chapters.

Basic vs. Applied Research

Definitions of these terms vary. *Basic research* can be characterized as that which attempts to determine or establish fundamental facts and relationships within a discipline or field of study. Andrew and Hildebrand (1982) define applied research as "research undertaken specifically for the purpose of obtaining information to help resolve a particular problem" (p. 3). Applied research has a definite purpose (p. x). A major distinction between basic and applied research is the focus of its application. Basic research is conducted with relatively little emphasis on its applications to "real-world" policy and management issues, while applied research is conducted with those issues at the forefront. This classification is of limited usefulness because it lacks clarity in its distinctions. For example, some basic research is conducted to develop a method, measurement, or theory so that an applied problem can be addressed. Is that research "basic" or "applied"? There is obviously more than one defensible position on the question.

The attempt to distinguish between basic and applied research may have its roots in attempts to classify basic and applied science. The premise of this distinction is that basic science ignores applications to current societal problems, while applied science focuses largely on those problems, sometimes without due concern for the underlying causal forces and relationships. This distinction is counterproductive to the extent that it causes scientists and researchers to limit their perspective of the relevance of their work. Nobel Prize recipient Herbert Simon (1979, p. 494) addressed the somewhat artificial distinction as follows:

It is a vulgar fallacy to suppose that scientific inquiry cannot be fundamental if it threatens to become useful, or if it arises in response to problems posed by the everyday world. The real world, in fact, is perhaps the most fertile of sources of good research questions calling for basic scientific inquiry.

Disciplinary, Subject-Matter, and Problem-Solving Research

This classification of types of research as described by Johnson (1986b) is perhaps the most comprehensive and useful one developed to date; reading of his treatment of the classification is important for the serious student of methodology. Breimyer (1991, p. 247) likewise recognizes the prominence of Johnson's classification con-

tribution to the study of research methodology. Figure 2.1 provides a characterization of the overlap between the basic/applied research classification and Johnson's disciplinary/subject-matter/problem-solving research classification.

By way of general description, *disciplinary research* is that which is "designed to improve a discipline" (Johnson, 1986b, p. 12). It dwells on theories, fundamental relationships, and analytical procedures and techniques within the discipline. Disciplinary research in economics is the research for which the intended users are other economists. While purely disciplinary research is not the

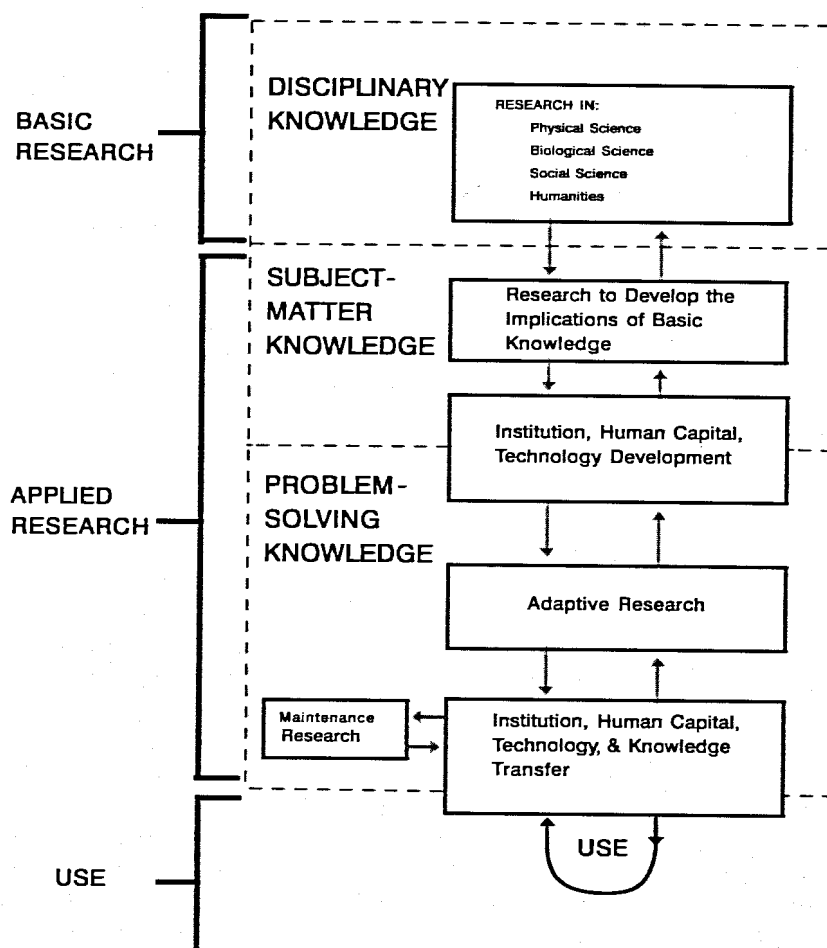


Figure 2.1. Relationship among research, knowledge, and use (Bonnen, 1986).

total research activity within economics, it is of utmost importance because it provides the conceptual and analytical base for the other types of economic research. Consequently, disciplinary research is synergistic and complementary with subject-matter and problem-solving (multidisciplinary) research; disciplinary research provides the foundations for much of the more applied research, while the applied research often reveals disciplinary shortcomings that need further disciplinary research (Johnson, 1991, p. 188).

Subject-matter research is "research on a subject of interest to a set of decision makers facing a set of practical problems" (Johnson, 1986b, p. 12). This type of research tends to follow subject-matter boundaries within a discipline—for example, resource economics, production economics, labor economics—and is inherently multidisciplinary because it draws information and ideas from other disciplines. Subject-matter research provides policy makers with the concepts and knowledge with which to make decisions about the sets of problems they address. Subject-matter research in economics commonly involves integrating subject-matter or issue knowledge from other disciplines such as engineering, sociology, or chemistry, with disciplinary (economic) concepts to draw implications about probable effects of proposed or possible actions. The distinctions between disciplinary and subject-matter research can be illustrated with the following examples. Medicine, a subject-matter field, utilizes the disciplines of biology, psychology, and chemistry (as well as other disciplines); management uses the disciplines of economics and psychology; engineering uses physics, mathematics, chemistry, and other disciplines. Subject-matter research is a cornerstone in economics in that it represents the direct application of economics to contemporary issues. Many economists consider subject-matter research to be directly relevant to current social and public issues and to be the primary source of policy applications for the discipline.

Other disciplines may be becoming more focused on subject matter. For example, biochemistry has adapted or integrated biological concepts and knowledge with chemistry concepts and knowledge, driven by a concern to develop useful, multidisciplinary knowledge. Bonnen (1986) goes further to suggest that this increased multidisciplinary focus is driven by public dissatisfaction with disciplinary research of unknown relevance (i.e., without specific purpose) and the corresponding shifts in research funding. He states that "to practice at the cutting edge in almost any discipline today, even in the biological and physical sciences, requires not only command of a discipline but also of major components of knowledge from related disciplines well beyond mathematics and statistics" (p. 1073). These types of subject-matter-driven integrations sometimes evolve into new disciplines. We may be seeing biochemistry evolving toward a distinct disciplinary status, even though its focus is largely subject-matter-oriented. All engineering fields (civil, electrical, industrial, petroleum, textile, etc.) are subject-matter-oriented and multidisciplinary. They draw from the theoretical realms of the "basic" sciences and mathematics and address problems defined in part by

external (often industry) parties. Subject-matter areas in colleges of agriculture (e.g., agronomy, animal science, horticulture) depend on their "basic" parent disciplines (botany, zoology, chemistry, etc.).

Economics is a separate discipline because it has its separate, distinct body of theory and empirical knowledge. The subject-matter areas using economics are inherently multidisciplinary. For example, consumer economics draws from psychology, natural resource economics from biology, and economic policy from political science. The various subject-matter areas are dependent on a common disciplinary base.² Thus, while economics is a separate discipline, much of what we eventually do with it—its applications—becomes multidisciplinary subject-matter work.

Problem-solving research is designed to solve a specific problem for a specific decision maker (Johnson, 1986b, p. 13). As is subject-matter research, it is inherently multidisciplinary because the problems that individual decision makers face invariably span disciplinary interests. Problem-solving research has a heavy focus on prescriptive knowledge (and it uses both positivistic knowledge and knowledge of values, which will be defined and discussed in chapter 3). Problem-solving research is more directly linked to a particular decision process than is subject-matter research. Problem-solving research is not more problem-oriented than the other types (all research is directed to a defined problem), but the problems it addresses are more oriented to specific, individual applications.

Problem-solving research often results in prescriptions or recommendations on decisions or actions. There is a difference of opinion on the delineation between problem-solving *research* and other problem-solving activities. Some view the problem-solving research activity as extending into the researcher being in the decision and implementation phases of the problem-solving process. The perspective advocated here is that the problem-solving research process stops short of implementation involvement. However, the distinction may not be critical in most respects. While the researcher may not make the decision or implement it, the prescriptions or recommendations become close to those activities.

Problem-solving research is holistic—it uses all information that is relevant to solving the specific problem at hand—while disciplinary research is reductionist (Johnson, 1987, p. 88), focusing only on the disciplinary content of the problem, often reducing the other components to a set of assumptions. Subject-matter research reaches across disciplines but is not focused on specific decision issues, and is therefore less holistic than problem-solving research. Disciplinary research is generally the most durable because the fundamental knowledge tends to maintain itself over a long period, while problem-solving research is generally the least durable because the situations addressed are specific and the least generalizable (Johnson, 1986b, ch. 16). Johnson (1986b) also notes that the three types of research must be judged by different criteria and perhaps from different philosophical perspectives. Johnson (1997, p. 261) states the case as follows:

When the object is to generate disciplinary knowledge, rigor can be regarded as disciplined careful efforts to improve the theoretical and empirical knowledge and techniques of an unambiguously defined discipline. . . . Multidisciplinary problem-solving work is often appraised using standards more appropriate for disciplinary work . . . (and) often foolishly denigrated . . . despite objectivity and care in carrying it out.

One way of synthesizing the fundamental differences, and some of the conflicts, between disciplinary and multidisciplinary perspectives with this classification system for research is as follows:

Those with responsibility for solving a problem or resolving an issue tend to view disciplinarians as potential consultants or advisors. Disciplinary knowledge becomes something to put in a multidisciplinary information retrieval system. . . . Until disciplinarians learn to suppress their disciplinary chauvinism, they are poorly qualified to administer problem-solving and issue-oriented investigations or to be decision makers. Disciplinarians and applied disciplinary knowledge by themselves typically cannot solve a problem or resolve an issue because of their specialized natures and because disciplines neglect interdependencies among the kinds of knowledge generated in different disciplines that are crucial in addressing the problem or issue at hand. (Johnson, 1997, pp. 258–259)

Analytical vs. Descriptive Research

Descriptive research may be characterized as simply the attempt to determine, describe, or identify what is, while *analytical research* attempts to establish why it is that way or how it came to be. The intent of descriptive research is often synthesis rather than analysis. Synthesis, or description, attempts to pull knowledge or information together and decipher its logically plausible connections (synthesize) rather than take the information apart to examine why and how it came to be (analyze). All disciplines generally engage in both types, although there may be differences in relative emphasis among disciplines. An example of descriptive research in economics is that which establishes the structure of an industry—number, size, location, and so on of firms within an industry. The corresponding analytical research on that same topic might be to determine the behavior of that industry in terms of pricing and other competitive behavior.

Economists see their discipline as having a pronounced analytical focus. Even so, we make extensive use of descriptive analysis in our research. We often employ descriptive analysis in the process of understanding, explaining, and documenting the problems on which we conduct research. We typically do this by gathering information and data, then putting them together in a way that it “tells a story” about the research problem or issue. An example is using data to show how real income of a group has changed over time, possibly in relation to the cost of

living for that same group. The data may be used to demonstrate a discrepancy between their movements and justify the need for more information on why the incomes and costs have moved as they have. Another way that descriptive research is widely used in economics is in subject-matter textbooks. While these textbooks may report results of analytical research, they tend to be based largely on making sense of the mass of information on the subject.

METHODOLOGY DEFINED AND DESCRIBED

The definition of methodology and the differentiation between methodology and method are matters on which reasonable people may disagree. The two terms are used interchangeably in many contexts. Rather than engage in a debate, methodology and methods are defined and delineated as they are used in this book.

Authorities are somewhat consistent in their definitions of methodology. Runes (1983, p. 212), in his *Dictionary of Philosophy*, defines methodology as "The systematic analysis and organization of the rational and experimental principles and processes which must guide a scientific inquiry. Also called scientific method. Thus, methodology is a generic term exemplified in the specific method of each science." *Webster's Collegiate Dictionary* (1977) defines methodology as "a body of methods, rules, and postulates employed by a discipline; . . . the analysis of the principles or procedures of inquiry in a particular field." Similarly, *Webster's New World Dictionary* (1968) defines it as "the science of method, or orderly arrangement; specifically the branch of logic concerned with the application of the principles of reasoning to scientific and philosophical inquiry . . . a system of method, as in any particular science."

Definitions of method are considerably more varied. *Webster's* (1977) definitions of method include "a way, technique, or process of or for doing something," and *Webster's* (1968) definitions include "a way of doing anything; mode; procedure; process; especially, a regular, orderly, definite procedure or way of teaching, investigating, etc." Breimyer (1991, p. 248) maintains that the term *method* is ill defined.

Using these definitions, the term *methodology* in this book refers to *the study of the general approach to inquiry in a given field*. Thus, research methodology in economics is the study of the general approach to research in economics. The term *method* is used to refer to *specific techniques, tools, or procedures applied to achieve a given objective*. This use of the term is consistent with that of the National Academy of Sciences' Committee on the Conduct of Science, which stated:

Throughout the history of science, some philosophers and scientists have sought to describe a single systematic method that can be used to generate scientific knowledge. Some scientists may believe in such a picture of themselves and their work, but carrying this approach into practice is impossible.

Rather than following a single scientific method, scientists use a body of methods particular to their work. . . . these methods include all of the techniques and principles that scientists apply in their work and in their dealings with other scientists. (Ayala et al., 1989, p. 2)

Research methods in economics include various versions of regression analysis, mathematical analysis, operations research techniques, surveys, gathering data, use of selected theoretical constructions, and other procedures, including combinations of techniques. Using this classification, *methods* and *methodology* are not interchangeable; methods are a portion of the concern of methodology.

These definitions may encounter some objections from some authors of journal articles, particularly in recent years. It has become commonplace to use the term methodology to describe, for example, specific analytical techniques such as Markov chains or certain econometric methods. However, if we are to apply rigor to the meanings of terms as we apply them in the study of research, definitions such as those above are necessary. Paarlberg (1963, p. 1386) offers the view that many economists "use the word methodology when they are concerned neither with philosophy nor with logic, but simply with method. The three syllables are added to convey prestige."

This point is also addressed by Machlup (1978). He states that "language pollution . . . has led to a vulgar use of the word methodology in a sense violating all philosophical traditions." He attributes this pollution to people being uninformed about meanings and too lazy to be precise; in this case adopting methodology as a synonym for method. Machlup states:

The propensity of many people to substitute technical terms for ordinary, well-understood words from the vocabulary of everyday speech has long been an effective force in the development of language, often leading to its enrichment, but sometimes to its pollution. The adoption of the term *methodology* by people ignorant of its original and proper meaning is a case of language pollution, probably of the irreparable sort, because we now find the word more often misapplied than not. (p. 6)

We also should make a clear distinction between research methodology in economics and economic methodology. *Research methodology in economics* is the study of the approach to research in the discipline of economics. *Economic methodology* is the study of the general approach to economic reasoning and validation of economic concepts. Economic methodology relates to the disciplinary basis (reasoning) of economics. Research methodology focuses on the process of developing information and knowledge, which may provide application knowledge and/or additional disciplinary understanding.

Boland (1982, p. 1) defines economic methodology as the view of the relationship between economic theories and methods of reaching conclusions about

the nature of the real world. Hausman (1989, p. 115) states that "economic methodology is concerned mainly with questions of theory confirmation or disconfirmation or empirical theory choice." Blaug (1980, p. xi) presents economic methodology as "philosophy of science applied to economics." The explanations of economists consist of theories. Economic methodology deals with questions of

what is the structure of these theories, and in particular, what is the relationship between assumptions and predictive implications in economic theories? If economists validate their theories by invoking factual evidence, is that evidence pertinent only to the predictive implications of theories, or to the assumptions of theories, or both? (Blaug, 1980, p. xii)

Other useful references on economic methodology include Caldwell (1982) and Stewart (1979).

While economic methodology and research methodology in economics are different, they are not independent. They depend on one another in much the same way as do science and research, which are discussed in chapter 3. Economic reasoning or logic and the processes by which we pursue economic knowledge is of necessity an integral part of the economic research process. Conversely, the ways we conduct research affects the body of accumulated economic knowledge and concepts. Yet if we do not mentally separate them, we risk the error of mistaking, for example, the process of selecting an appropriate model or concept for the process of evaluating whether the model or concept applies. Taking the position that a theory is applicable in a given situation and testing its applicability in that same situation are *not* the same; the former is likely a prelude to or a part of the latter.

In some respects, research methodology might be viewed as a subset of economic methodology because the research process is a means for developing and evaluating economic theory. However, that perspective may be misleading. The two methodologies are more accurately viewed as symbiotic—each depends on, and supports, the other.

This symbiotic relationship can be illustrated by drawing from research on the general topic of hedonic or characteristic price analysis. The theoretical basis for hedonic/characteristic price analysis lies with the works of Lancaster (1966) and Rosen (1974), with several notable prior and subsequent works. Research methodology is employed in conducting specific and general exploration of the valuation of characteristics or attributes embodied in goods. Research methodology is concerned with how that analytical process is undertaken—developing models (based on theory, which is, in turn, related to economic methodology) for analyzing the relationships and parameters, selecting and obtaining data with which to estimate relationships and test hypotheses, evaluating implications of the relationships, and so on. What is learned, in turn, may provide additional insight into how markets work to value attributes or characteristics: that is, how

we *think about* markets and their operation (our theories about markets). The new insight, understanding, and reasoning (our slightly expanded and/or altered theoretical base) then provide an expanded base of theory on which additional research can be built.

THE PROCESS OF RESEARCH

As indicated earlier, research involves or constitutes a process. While the specifics of the way in which the process may be implemented or applied has infinite variations, figure 2.2 is a stylized representation of the process. The process is initiated with a question or problem—the researcher's perceived need for knowledge about something. The question or problem can come from various sources, and we can make some generalizations about the types of research that are more likely to ensue depending on the source(s) of the problem(s). As with all generalizations, they are fallible. Questions exclusively within the discipline are likely to be theoretical or procedural questions that lead to disciplinary research. Questions coming in total or in part from related disciplines, the government, or public-interest groups are more likely to lead to subject-matter or problem-solving research. Questions influenced by firms or industry groups are apt to be problem solving in their orientation.

Given the question, problem, or issue, the next step is to formulate goals or objectives to deal with the problem. These objectives, as with the problem, are

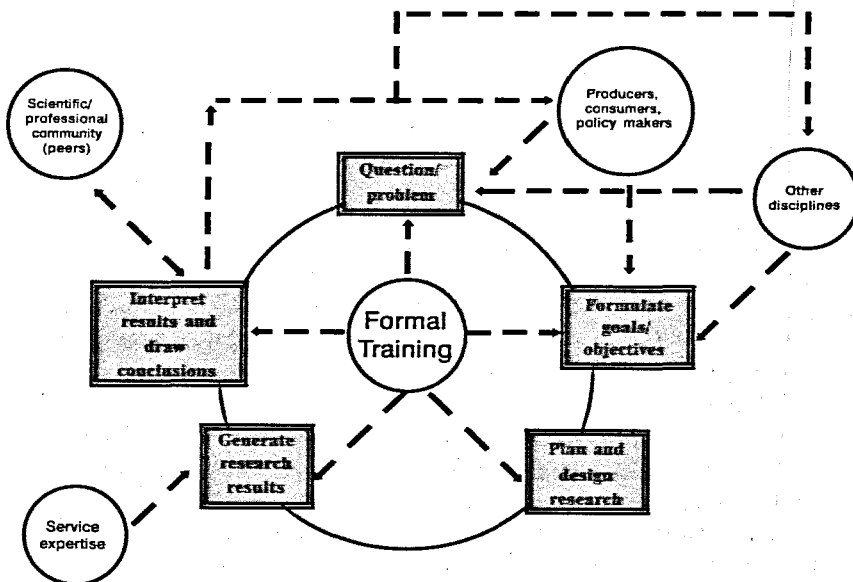


Figure 2.2. Schematic of research process.

likely to be influenced by the research clientele—other economists, industry, the public, or other disciplines. As part of a *research* process, these research goals relate to obtaining knowledge, information, or understanding about the problem/issue/question. Research objectives or goals may include deriving prescriptions and/or recommendations, especially when problem-solving research is involved.

The third step in the process, research design, becomes more isolated from other parties when it involves specialized techniques. It is, however, influenced by solicited specialized expertise from within or outside the discipline. For example, we might seek theoretical or modeling insight from colleagues, such as mathematicians or statisticians, who are more experienced or specialize in certain topics. The research design is formulated to achieve the research objectives, may contain many elements, and is the basis for accepting research findings. The reliability of results from a given research endeavor depends on the validity of the research design, never on its findings.

The fourth step, that of generating or producing results, is often a more “mechanical” step in comparison to the steps that precede it, especially in empirically oriented economic research. For example, this may be the step in which we generate parameter estimates by using statistical techniques from econometric model(s) formulated in the research design, or make projections using mathematical simulation model(s) constructed in the research design. In empirical research, service expertise might include computer-programming input. If the research is not empirical in nature, it is the implementation of whatever research plan was formulated in the previous step and is likely not “mechanical.” In theoretical disciplinary research, this step may consist largely of the application of deductive reasoning to the premises and situation being analyzed.

Once results are obtained, they must be interpreted and analyzed themselves (step 5). This is the step in which we look at implications, draw conclusions, and perhaps offer prescriptions. For example, empirical results from an econometric model are analyzed and interpreted to determine what they show in terms of policy and management applications. We may derive estimates of demand and supply elasticities from an econometric market model, then examine the implications of the elasticities and compare the elasticity estimates to those estimated from other studies. Step 5 is the one most often overlooked or underestimated by students as they are being initiated to the research process. They often see the empirical results as the end product of the process. Actually, the interpretation of those results may be more challenging, and more important, than the process of obtaining the empirical estimates.

Once step 5 is completed, results are typically disseminated. Dissemination within the discipline occurs through disciplinary and subject-matter journals, professional meeting presentations, and other types of research reports. Dissemination to the public occurs through research reports, presentations, and occasionally public media coverage. Dissemination to related disciplines occurs through

research reports, presentations, and multidisciplinary journal papers. Dissemination within a corporation or government agency may consist of an internal research report and/or an oral report. Presentation to the different audiences must vary because of differences in background and expertise of the clientele. The different types of audiences are each important in their own way and maybe for different reasons.

The solid line between step 5 and step 1 (figure 2.2) closes the loop and makes the research process circular. Getting answers to some questions always leads to other questions. The more we understand about a given phenomenon, the more clearly we see the particular aspects of it that we do not yet understand. The ensuing questions may be in the same vein as the research questions originally posed, or they may be more peripheral to the original question. It is this characteristic, however, that evidences the need for the process to be planned and organized. Unless we impose some limits or bounds on an individual research effort, the tendency is for it to be a continuous question-answer-question progression. By imposing some closure on an individual effort, the process becomes one of a succession of research projects, each potentially linking to others and the accumulation of knowledge about the topic. While slower than we may prefer, the process is at least systematic. By periodically disseminating procedures, results, and conclusions, information is released to others working on the same or similar topics. The subsequent feedback from extended research users, represented by the dotted lines between the public policy makers (figure 2.2), is an important part of the loop, especially with problem-solving and subject-matter research. Users react to research output, then provide additional questions and issues for research.

CREATIVITY IN THE RESEARCH PROCESS

Research is a creative process. This is a point that cannot be overemphasized. Brewster (1970, pp. 234–235) states that

there is much to the research quest that cannot be reduced to systematic rules. . . . research includes far more than mere logic, whether deductive or inductive. It includes insight, genius, groping, pondering—"sense" which can't be boxed up in any formalized procedures. The logic we can teach; the art we cannot.

The research process requires, or at least works best with, imagination, initiative, a degree of intuition, and a large dose of curiosity. There is no "magic formula" for accomplishing or producing good economic research. If it were that simple, anyone could do it. There may be, and often are, different ways to accomplish a given research mission or objective. We may occasionally even conduct research to determine which approach is more effective or efficient.

Creativity is a somewhat vague concept on which written material seems to provide no clear definition. Creativity seems somewhat like "art"—we cannot define it, but we know it when we see it. For our purposes, it is defined as the generation and implementation of new ideas. One reason for confusion about creativity is that there are different types of creativity (Stuller, 1982, pp. 38, 39):

A. *Imaginative creativity*. This is creativity through imagination—purely imaginary (non-factual) mental constructions. This type of creativity is usually associated with painters and writers. However, economists engage in imaginative creativity when we, e.g., formulate hypothetical relationships or causal linkages which we then proceed to evaluate and test.

B. *Natural creativity*. This is creativity through natural physical or mental ability and is usually exhibited by performance. This type of creativity is associated with dancers, musicians, and athletes. Natural creativity has little to do with economics unless it is extended to the sphere of mental activity. Perhaps some have a greater capacity for economic logic than others.

C. *Prescriptive creativity*. This is creativity through development of new possibilities. It is stimulated by "what if . . ." questions, and tends to be associated with philosophy and religion. Economists engage in prescriptive creativity in, e.g., arriving at public policy prescriptions and guidelines.

D. *Applied creativity*. This is creativity in the form of application of concepts to solve problems. It is not associated with any field in particular, but is an activity in any "problem solving" discipline. Applied creativity is an integral part of economists' activities because our activities are oriented largely to addressing societal and managerial problems.

E. *Theoretical creativity*. This is creativity through the development of relationships between things; it results in abstract concepts. Theoretical creativity is a part of most academic disciplines, although not necessarily all fields of endeavor. Economists draw on theoretical creativity in their disciplinary theoretical work.

The degree to which applied and theoretical creativity can be separated, at least in economics, may be questioned. For example, Bonnen (1986, p. 1072) presents the view that "Much of the creativity in any discipline comes from the intellectual stimulation of confronting disciplinary knowledge with the test of application."

Intuition and imagination may be important parts of all types of creativity. Creativity is rarely achieved without tenacity and perseverance, and the evidence of creativity is not often realized without work. Psychologists believe that creativity is an acquired ability to a large extent—it can be learned (Gilbert, 1986), which would appear to be consistent with the view that creativity can be acquired. The American Economic Association's Commission on Graduate Education in Economics expressed concern that graduate programs may be stifling creativity

by overemphasizing acquisition of technical skills in the early years of graduate programs. The commission recognized that "creativity relevant to economic research is multidimensional. It can be manifested by posing a new question, by reformulating an existing problem, by finding new data sources with which to test a proposition, by developing a new model, or by extending mathematical or statistical techniques" (Krueger et al., 1991, p. 1049).

We lack prescriptions on how to be creative, much the same as there are no prescriptions for how to be "talented" or how to be an effective communicator. There are, however, some things you can practice that will foster creativity (as there are certain guidelines that will help you become an effective communicator). Stuller (1982) provides a set of suggestions for fostering creativity, and Ladd (1987) identifies conditions conducive to creativity. These provide the basis for the following guidelines for helping you to be creative:

A. Gather and use previously developed knowledge; thorough preparation stimulates unconscious thought processes.

B. Exchange ideas with other people, especially your colleagues.

C. Capture your intuition by applying deductive logic. When your intuition suggests something, make note of it before it escapes, then deal with it on a logical, analytical level.

D. Cultivate the ability to look at things in alternative ways. Diversity of interests and experiences, as well as maintaining a degree of doubt about almost everything, helps to develop this ability.

E. Question or challenge assumptions.

F. Practice searching for patterns or relationships among occurrences or things. This includes Ladd's sensitivity to similarities (perception of a link between things not previously connected) and using combinations of the above practices.

G. Do not be afraid to take risks; little can be gained if you are not willing to explore. Ladd refers to this as developing a venturesome attitude. A statement (source unknown) may illustrate the point: moderation is fine for parents and politicians, but science, literature, and the arts need boldness.

H. Cultivate a degree of tolerance for uncertainty. When you know things that are contradictory, give yourself time to ponder them and seek a resolution; a temporary suspension of judgment may allow the unconscious mind to reconcile the contradiction.

I. Allow your curiosity—your intense desire to know or understand—to build a creative tension.

J. When you are unable to resolve a troublesome problem, try abandoning it, at least in your conscious mind, for a while. This may let the unconscious mind work on it and provide new insight when you return to it with your conscious mind later.

K. Discipline yourself to write down your thoughts. There is something about having to express your thoughts in writing, so that they are clear to someone else, that fosters your own process of mental discovery. Ladd (1987, p. 77) states, "Many of my intuitions come to me when I am writing. It frequently is true that I do not know what I think until I write it." More will be said about the role of writing in chapter 5.

L. Create situations in which you have freedom from distraction. Having some protected time in which to concentrate and contemplate may be necessary for creative thinking.

Psychology provides us with the understanding that the two hemispheres of the brain are somewhat specialized. The left side is more dominant in functions involving abstract reasoning and language, while the right side is more dominant in visual and perception skills, intuition, and expression of emotion (Levy, 1985; Buffington, 1986). The left hemisphere processes information sequentially, while the right hemisphere processes information simultaneously (Ladd, 1987, p. 29). Each side is capable of functioning independently of the other and they work differently and come up with different answers. Austin (1978, p. 139) explains that the left hemisphere "proceeds, piecemeal, to examine the irregular bark on each tree . . . [while the] right hemisphere grasps in one sweep the shape of the whole forest, relates it . . . to the contours of the near landscape, then to the line of the horizon." However, creative thinking involves a free flow of messages between conscious and unconscious (Ladd, 1987, p. 27); creativity is produced by integrating the abilities of both sides. The interaction creates a special mental procedure that exceeds the capacity of the independent parts.

This spark of creativity, if you can find it, may be the difference between satisfactory and outstanding research. Perhaps good (i.e., useful) research may be done without creativity, and creativity does not guarantee good research. However, the most innovative research probably requires creative thinking.

SUMMARY

Research entails different activities and approaches among disciplines, and the public's comprehension of economic research may arguably be more limited than it is of research in fields such as medicine, engineering, and chemistry. This chapter establishes some definitions of essential terms, classifications of types of research, distinctions between sometimes confusing terms, and an overview of the research process in economics. The importance and role of creativity is emphasized.

Research is defined as the systematic approach to obtaining new and reliable knowledge. Activities such as accidental discovery, data collection, and reviewing

COLORADO STATE UNIVERSITY LIBRARY

published research do not, by themselves, constitute research, but each plays a part in research. Economic research is a structured, interactive, creative, circular process of searching for explanations to economic problems and phenomena. Research is presented under three different classifications: (1) basic and applied research; (2) disciplinary, subject-matter, and problem-solving research; and (3) analytical and descriptive research. The different classifications are each useful.

Research methodology—the study of the general approach to the research process in economics—is distinguished from research methods—particular techniques, tools, or procedures used to meet a set of research objectives. Also, research methodology in economics is distinguished from economic methodology. The process of research methodology in economics is described as being circular with certain steps or components. The steps consist of problem/question/issue specification, objective identification, design of methods and procedures, generation of results, and interpretation of results. The process typically identifies a new set of research problems.

The role of individual creativity and imagination in the process is emphasized, with clarification and examples provided on different types of creativity and how each may be relevant in economic research. Suggestions on how to foster creativity are also provided.

RECOMMENDED READING

Bloom et al. *Taxonomy of Educational Objectives*, pp. 201–207.

Hausman. “Economic Methodology in a Nutshell.”

Johnson. *Research Methodology for Economists*, pp. 1–7, 11–14.

Ladd. *Imagination in Research*. The entire book is recommended reading, but most especially parts I and II, pp. 1–94.

Machlup. *Methodology of Economics*, pp. 5–10.

SUGGESTED EXERCISES

1. Describe an example of each type of creativity identified in this chapter from your own experience.

2. Select three articles, each from a different recent issue of an economics journal. Classify each article according to the type(s) of research reported in the article and explain why you classify them as such.

3. Select three recent journal articles (they may be the same as in exercise 2 above). Is the term *methodology* used in each? If it is, is the term used as it has been defined here or is it used to describe a method or technique?

4. Of the techniques described to help foster creativity, which do you use regularly? Describe, in writing, how you employ them.

NOTES

1. This does not say that the outcomes are anticipated or that unplanned unsystematic discoveries do not occur. Any effort to discern the unknown may lead to unexpected knowledge (accidental discovery). But without a systematic approach, one is left groping in random directions rather than exploring along a planned avenue.

2. Agricultural economics is viewed as a separate discipline by some—usually agricultural—economists. Those who view it as a separate discipline seem to base the distinction on the fact that agricultural economics did not evolve directly from economics. Instead it adopted its parent discipline after its initial inception. The author's inclination is to view agricultural economics as a subject-matter area of economics, but of a somewhat different status than other subject-matter areas because of its leading role in the development of empirical quantitative analysis in economics. Much of the development and testing of empirical methods, which is central to scientific validation of theory (Eichner, 1986), came first in agricultural economics.